Lecture Notes: Introduction to Machine Learning

Supervised learning algorithms build a mathematical model of a set of data that contains both inputs and the desired outputs. Gradient descent is an optimization algorithm used to minimize the loss function by iteratively moving in the direction of steepest descent as defined by the negative gradient.

Lecture Notes: Neural Networks and Overfitting

Overfitting occurs when a machine learning model learns the training data too well, capturing noise and details that do not generalize to unseen data. Common regularization techniques to combat overfitting include L1 regularization, L2 weight decay, and Dropout layers during backpropagation training.